

```
import os
os.environ["OPENAI_API_KEY"] = "*****"
```

```
#installation
!pip install langchain chromadb openai tiktoken pypdf langchain_openai langchain-community

Downloading langchain-classic-1.0.0-py3-none-any.whl (110 kB)
1.0/1.0 MB 51.4 MB/s eta 0:00:00
Downloading onnxruntime-1.27.0-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (18.7 MB)
18.7/18.7 MB 77.4 MB/s eta 0:00:00
Downloading opentelemetry_exporter_otlp_proto_grpc-1.43.0-py3-none-any.whl (19 kB)
Downloading opentelemetry_exporter_otlp_proto_common-1.43.0-py3-none-any.whl (17 kB)
Downloading opentelemetry_proto-1.43.0-py3-none-any.whl (72 kB)
72.5/72.5 kB 5.4 MB/s eta 0:00:00
Downloading opentelemetry_sdk-1.43.0-py3-none-any.whl (178 kB)
178.9/178.9 kB 13.6 MB/s eta 0:00:00
Downloading opentelemetry_api-1.43.0-py3-none-any.whl (61 kB)
61.9/61.9 kB 4.7 MB/s eta 0:00:00
Downloading opentelemetry_semantic_conventions-0.64b0-py3-none-any.whl (203 kB)
203.7/203.7 kB 14.8 MB/s eta 0:00:00
Downloading overrides-7.7.0-py3-none-any.whl (17 kB)
Downloading pybase64-1.4.3-cp312-cp312-manylinux1_x86_64.manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_5_x86_64.whl (
71.6/71.6 kB 5.5 MB/s eta 0:00:00
Downloading pypika-0.51.1-py2.py3-none-any.whl (60 kB)
60.6/60.6 kB 4.1 MB/s eta 0:00:00
Downloading requests-2.34.2-py3-none-any.whl (73 kB)
73.1/73.1 kB 5.5 MB/s eta 0:00:00
Downloading durationpy-0.10-py3-none-any.whl (3.9 kB)
Downloading langchain_text_splitters-1.1.2-py3-none-any.whl (35 kB)
Downloading pyproject_hooks-1.2.0-py3-none-any.whl (10 kB)
Installing collected packages: pypika, durationpy, requests, pyproject_hooks, pypdf, pybase64, overrides, opentelemetry-proto,
  Attempting uninstall: requests
    Found existing installation: requests 2.32.4
    Uninstalling requests-2.32.4:
      Successfully uninstalled requests-2.32.4
  Attempting uninstall: opentelemetry-proto
    Found existing installation: opentelemetry-proto 1.38.0
    Uninstalling opentelemetry-proto-1.38.0:
      Successfully uninstalled opentelemetry-proto-1.38.0
  Attempting uninstall: opentelemetry-api
    Found existing installation: opentelemetry-api 1.38.0
    Uninstalling opentelemetry-api-1.38.0:
      Successfully uninstalled opentelemetry-api-1.38.0
  Attempting uninstall: opentelemetry-semantic-conventions
    Found existing installation: opentelemetry-semantic-conventions 0.59b0
    Uninstalling opentelemetry-semantic-conventions-0.59b0:
      Successfully uninstalled opentelemetry-semantic-conventions-0.59b0
  Attempting uninstall: opentelemetry-exporter-otlp-proto-common
    Found existing installation: opentelemetry-exporter-otlp-proto-common 1.38.0
    Uninstalling opentelemetry-exporter-otlp-proto-common-1.38.0:
      Successfully uninstalled opentelemetry-exporter-otlp-proto-common-1.38.0
  Attempting uninstall: opentelemetry-sdk
    Found existing installation: opentelemetry-sdk 1.38.0
    Uninstalling opentelemetry-sdk-1.38.0:
      Successfully uninstalled opentelemetry-sdk-1.38.0
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is th
google-colab 1.0.0 requires requests==2.32.4, but you have requests 2.34.2 which is incompatible.
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-exporter-otlp-proto-common==1.38.0, but you have openteler
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-proto==1.38.0, but you have opentelemetry-proto 1.43.0 wh
opentelemetry-exporter-otlp-proto-http 1.38.0 requires opentelemetry-sdk~=1.38.0, but you have opentelemetry-sdk 1.43.0 wh
opentelemetry-exporter-gcp-logging 1.11.0a0 requires opentelemetry-sdk<1.39.0,>=1.35.0, but you have opentelemetry-sdk 1.43.0
google-adk 1.29.0 requires opentelemetry-api<1.39.0,>=1.36.0, but you have opentelemetry-api 1.43.0 which is incompatible.
google-adk 1.29.0 requires opentelemetry-sdk<1.39.0,>=1.36.0, but you have opentelemetry-sdk 1.43.0 which is incompatible.
Successfully installed bcrypt-5.0.0 build-1.5.0 chromadb-1.5.9 durationpy-0.10 kubernetes-36.0.2 langchain-classic-1.0.8 langc
```

```
!pip install langchain-chroma
```

```

Requirement already satisfied: websocket-client!=0.40.0,!0.41.*,!=0.42.*,>=0.32.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from kubernetes==28.1.0->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: requests-oauthlib in /usr/local/lib/python3.12/dist-packages (from kubernetes==28.1.0->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: urllib3!=2.6.0,>=1.24.2 in /usr/local/lib/python3.12/dist-packages (from kubernetes==28.1.0->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: durationpy>=0.7 in /usr/local/lib/python3.12/dist-packages (from kubernetes==28.1.0->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: aiohttp<4.0.0,>=3.13.5 in /usr/local/lib/python3.12/dist-packages (from kubernetes==28.1.0->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: requests-toolbelt>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45->langchain)
Requirement already satisfied: websockets>=15.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45->langchain)
Requirement already satisfied: xxhash>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45->langchain)
Requirement already satisfied: zstandard>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45->langchain)
Requirement already satisfied: flatbuffers in /usr/local/lib/python3.12/dist-packages (from onnxruntime==1.14.1->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: protobuf>=4.25.8 in /usr/local/lib/python3.12/dist-packages (from onnxruntime==1.14.1->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: googleapis-common-protos~=1.57 in /usr/local/lib/python3.12/dist-packages (from opentelemetry-exporter-otlp-proto-common==1.43.0->opentelemetry-exporter-otlp-proto-grpc==1.43.0->opentelemetry-exporter-otlp-proto-http==1.43.0->opentelemetry-exporter-otlp-proto-common==1.43.0->opentelemetry-exporter-otlp-proto-common==1.43.0)
Requirement already satisfied: opentelemetry-exporter-otlp-proto-common==1.43.0 in /usr/local/lib/python3.12/dist-packages (from opentelemetry-exporter-otlp-proto-grpc==1.43.0->opentelemetry-exporter-otlp-proto-http==1.43.0->opentelemetry-exporter-otlp-proto-common==1.43.0)
Requirement already satisfied: opentelemetry-proto==1.43.0 in /usr/local/lib/python3.12/dist-packages (from opentelemetry-exporter-otlp-proto-grpc==1.43.0->opentelemetry-exporter-otlp-proto-http==1.43.0->opentelemetry-exporter-otlp-proto-common==1.43.0)
Requirement already satisfied: opentelemetry-semantic-conventions==0.64b0 in /usr/local/lib/python3.12/dist-packages (from opentelemetry-exporter-otlp-proto-grpc==1.43.0->opentelemetry-exporter-otlp-proto-http==1.43.0->opentelemetry-exporter-otlp-proto-common==1.43.0)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic==2.0->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic==2.0->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic==2.0->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: python-dotenv>=0.21.0 in /usr/local/lib/python3.12/dist-packages (from pydantic-settings==2.0.0->pydantic==2.0->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=10.11.0->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=10.11.0->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: huggingface-hub<2.0,>=0.16.4 in /usr/local/lib/python3.12/dist-packages (from tokenizers==0.13.0->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer==0.9.0->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: shellingham==1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer==0.9.0->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer==0.9.0->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: httpxtools>=0.8.0 in /usr/local/lib/python3.12/dist-packages (from uvicorn[standard]>=0.18.3->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: uvloop>=0.15.1 in /usr/local/lib/python3.12/dist-packages (from uvicorn[standard]>=0.18.3->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: watchfiles>=0.20 in /usr/local/lib/python3.12/dist-packages (from uvicorn[standard]>=0.18.3->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.13.5->kubernetes)
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.13.5->kubernetes)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.13.5->kubernetes)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.13.5->kubernetes)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.13.5->kubernetes)
Requirement already satisfied: yarll<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp<4.0.0,>=3.13.5->kubernetes)
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0.16.4->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0.16.4->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.5.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=0.16.4->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=10.11.0->chromadb<2.0.0,>=1.0.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->kubernetes)
Requirement already satisfied: oauthlib>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from requests-oauthlib->kubernetes)
Downloading langchain_chroma-1.1.0-py3-none-any.whl (12 kB)
Installing collected packages: langchain-chroma
Successfully installed langchain-chroma-1.1.0

```

```

from langchain_openai import OpenAIEmbeddings
from langchain_chroma import Chroma

```

```

from langchain_core.documents import Document

# Create LangChain documents for IPL players

doc1 = Document(
    page_content="Virat Kohli is one of the most successful and consistent batsmen in IPL history. Known for his aggressive batting style, he has scored over 7000 runs in IPL. He is a right-handed batsman and a right-arm medium bowler."
    metadata={"team": "Royal Challengers Bangalore"}
)
doc2 = Document(
    page_content="Rohit Sharma is the most successful captain in IPL history, leading Mumbai Indians to five titles. He's a right-handed batsman and a right-arm medium bowler."
    metadata={"team": "Mumbai Indians"}
)
doc3 = Document(
    page_content="MS Dhoni, famously known as Captain Cool, has led Chennai Super Kings to multiple IPL titles. His finishing skills are legendary. He is a right-handed batsman and a right-arm medium bowler."
    metadata={"team": "Chennai Super Kings"}
)
doc4 = Document(
    page_content="Jasprit Bumrah is considered one of the best fast bowlers in T20 cricket. Playing for Mumbai Indians, he has taken over 150 wickets in IPL. He is a left-arm fast bowler."
    metadata={"team": "Mumbai Indians"}
)
doc5 = Document(
    page_content="Ravindra Jadeja is a dynamic all-rounder who contributes with both bat and ball. Representing Chennai Super Kings, he has been a key player in their success. He is a left-handed batsman and a left-arm medium bowler."
    metadata={"team": "Chennai Super Kings"}
)

```

```
docs = [doc1, doc2, doc3, doc4, doc5]
```

```

#forming a vector store using chroma
vector_store = Chroma(

```

```
embedding_function=OpenAIEmbeddings(),
persist_directory="chroma-db",
collection_name='cricket_sample'
)
```

```
#add docs
vector_store.add_documents(docs)
```

```
# view documents
vector_store.get(include=['embeddings','documents', 'metadatas'])
```

```
# search documents
vector_store.similarity_search(
    query='Who among these are a bowler?',
    k=2
)
```

```
# search with similarity score
vector_store.similarity_search_with_score(
    query='Who among these are a bowler?',
    k=2
)
```

```
# meta-data filtering
vector_store.similarity_search_with_score(
    query="",
    filter={"team": "Chennai Super Kings"}
)
```

```
# update documents
updated_doc1 = Document(
    page_content="Virat Kohli, the former captain of Royal Challengers Bangalore (RCB), is renowned for his aggressive leadership",
    metadata={"team": "Royal Challengers Bangalore"}
)

vector_store.update_document(document_id='09a39dc6-3ba6-4ea7-927e-fdda591da5e4', document=updated_doc1)
```

```
# view documents
vector_store.get(include=['embeddings','documents', 'metadatas'])
```

```
# delete document
vector_store.delete(ids=['09a39dc6-3ba6-4ea7-927e-fdda591da5e4'])
```

```
# view documents
vector_store.get(include=['embeddings','documents', 'metadatas'])
```

